# From Text to Memes: A Progressive Pipeline for Misogyny Detection.

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## Abstract

Online misogyny is a serious and growing problem, but detecting it automatically is harder than it looks. Obvious insults are easy to catch, but subtler forms like gender stereotypes, backhanded compliments, or sarcastic emojis are much harder for machine learning models to handle. In this project, we build a progressive pipeline that tackles this challenge in three steps: starting from an emoji-aware RoBERTa model trained on EDOS and Hatemoji, extending it to Reddit posts via OCR fine-tuning, and finally combining text and images for meme classification with CLIP. We also conduct a detailed analysis of where and why models fail. Our results show that models detect explicit misogyny well (recall 0.81) but consistently miss implicit forms (recall 0.56). We also find that models are biased toward gendered words: a non-misogynistic sentence containing the word *female* gets flagged as misogynistic 23.8% of the time, compared to just 1.1% for sentences without any gendered terms. The multimodal model achieves the best overall performance with a macro F1 of 0.855 on the MAMI meme dataset.

**Keywords:** misogyny detection, hate speech, RoBERTa, CLIP, emojis, implicit sexism, multimodal.

## 1. Introduction

Social media platforms have made it easier than ever for people to express themselves, but this openness also comes with a darker side. Misogynistic content ranging from direct insults to subtle stereotypes is widespread online and has real consequences for the women who encounter it [1]. Automatically detecting this type of content is an important step toward making online spaces safer, but it is far from a solved problem.

The main difficulty is that misogyny does not always look obvious. A sentence like *"She's pretty smart for a girl"* is sexist, but it does not contain any explicit slurs. Similarly, adding a rolling-eyes emoji 🙄 to *"Women are so smart"* completely changes the meaning of an otherwise positive sentence, turning a compliment into sarcasm. Standard text classifiers that ignore emojis or rely on keyword matching are likely to miss these cases entirely.

We investigate three key questions in this project. First, how well do current transformer models handle *implicit* misogyny, as opposed to the more obvious explicit kind? Second, do these models exhibit systematic biases, for example flagging any sentence that mentions women as potentially misogynistic? Third, does adding extra information like emojis or images actually improve detection performance?

To answer these questions, we build a progressive pipeline with three steps. We start with an emoji-aware RoBERTa model fine-tuned on EDOS and Hatemoji, then adapt it to the Reddit domain via OCR fine-tuning, and finally combine it with CLIP for multimodal meme classification. In parallel, we conduct a detailed analysis of where and why models fail, including implicit vs explicit sexism detection, gender-term bias, and a synthetic diagnostic benchmark.

## 2. Related Works

**Transformer models for hate speech.** BERT [2] and its variants are the standard for text classification. Caselli et al. [3] retrained BERT on abusive Reddit content (HateBERT), and Liu et al. [4] proposed RoBERTa, a more robustly trained variant that outperforms BERT on most tasks.

**Sexism detection benchmarks.** Kirk et al. [5] built EDOS, a dataset of 20,000 posts with fine-grained sexism labels (threats, derogation, stereotyping, condescension). Models trained on EDOS do well on obvious sexism but struggle with subtler categories.

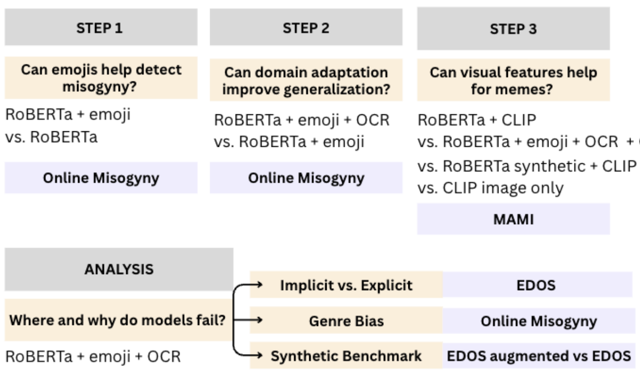
**Emoji-based hate speech.** Kirk et al. [6] showed that hate speech models break down when offensive meaning is conveyed through emojis, motivating the adversarially-generated Hatemoji dataset.

**Multimodal misogyny in memes.** Fersini et al. [7] created MAMI, the first large meme benchmark, showing that models combining image and text outperform text-only ones. Radford et al. [8] introduced CLIP, trained on 400M image-text pairs, widely used as a visual encoder.

**Gaps addressed by our work.** Most existing models are tested on data that looks very similar to what they were trained on, which makes their results look better than they actually are in practice. None of them specifically study whether models are biased toward gendered words, or how emojis interact with implicit sexism. In our project, we look at all of these issues.

## 3. Method

Our approach follows a progressive pipeline with three steps plus a dedicated analysis component.




**Step 1 — Emoji-aware RoBERTa.** We fine-tune `roberta-base` [4] on EDOS + Hatemoji (21,319 samples) and test on Online Misogyny (6,356 Reddit posts). Emojis are converted to text descriptions using `demojize` (e.g. 🙄 → `:face_with_rolling_eyes:`), preserving semantic meaning without modifying the model vocabulary. We apply balanced class weights ($w_c = N/2N_c$) and use a cross-domain setup for an honest measure of generalization. To isolate the contribution of emojis, we compare against a RoBERTa baseline trained on EDOS only.

**Step 2 — Domain adaptation via OCR fine-tuning.** The Online Misogyny dataset contains Reddit posts with associated images (e.g., screenshots, memes), which often include additional textual content. To capture this information, we apply OCR to extract text from images and concatenate it with the original post text.

We then fine-tune the Step 1 checkpoint on this OCR-enriched dataset (3 epochs, lr=$2 \times 10^{-5}$, batch size 32). This captures additional textual signals and comparison with Step 1 isolates the impact of OCR-based domain adaptation.

**Analysis — Where and why do models fail?** Without retraining, we evaluate model failures on the best Step 2 checkpoint by: (a) splitting the EDOS test set into explicit (threats, slurs, derogation) and implicit (stereotypes, backhanded compliments, condescension) categories using official labels, (b) measuring false positive rates on non-misogynistic text containing gendered terms, and (c) running 20 controlled emoji minimal pairs. Each pair consists of the same sentence with and without a semantically relevant emoji (e.g. *"Women are so smart"* vs *"Women are so smart 🙄"*), to isolate the effect of emojis on predictions. We also stress-test the model using a synthetic benchmark of 300 ChatGPT-generated examples (100 explicit, 100 implicit, 100 neutral/feminist) plus 50 emoji pairs.

**Step 3 — Multimodal model (RoBERTa + CLIP).** To classify memes, we combine RoBERTa (text encoder, `[CLS]` projected to 256 dimensions) with a frozen CLIP `ViT-B/32` [8] (image encoder, 512→256). CLIP is pretrained on large-scale image–text pairs and therefore already captures implicit multimodal interactions directly from meme images, making it a strong baseline even without explicit text modeling. We freeze CLIP because MAMI (∼10,000 examples) is too small to reliably fine-tune a model trained on 400M image–text pairs, which would likely lead to overfitting and degrade generalization. Instead, CLIP is used as a fixed visual feature extractor: text and image features are projected to a common space (256 dimensions each), concatenated (512 dimensions), and fed into a small classifier (512→128→2, ReLU). We use simple concatenation for stability and data efficiency, as cross-modal attention requires more data to train reliably. While CLIP captures implicit image–text interactions, combining it with RoBERTa enables explicit modeling of linguistic content, which can improve performance but may introduce redundancy. To isolate the contribution of the text encoder, we compare four variants on MAMI: CLIP only with all the previous steps fine-tuned RoBERTa models + CLIP. Training: MAMI [7], 3 epochs, lr=$2 \times 10^{-5}$, batch size 16, balanced class weights.

## 4. Validation

| Model | F1 | Test set |
|---|---|---|
| RoBERTa (no emoji) | 0.719 | Online Misog. |
| RoBERTa + emoji | 0.715 | Online Misog. |
| RoBERTa + emoji + OCR FT | **0.738** | Online Misog. |
| CLIP image only | 0.789 | MAMI |
| RoBERTa base + CLIP | 0.849 | MAMI |
| RoBERTa synthetic + CLIP | 0.849 | MAMI |
| RoBERTa + emoji + OCR FT + CLIP | **0.855** | MAMI |

*Table 1.* Macro F1 across all models and test sets.

**Step 1 — Effect of emoji tokenization.** Adding emoji tokenization slightly decreases F1 from 0.719 to 0.715 on Online Misogyny. This is partly explained by the absence of emojis in the test set (0% of samples), preventing the model from leveraging its added capability. The emoji component is better evaluated through controlled experiments (see Analysis).

**Step 2 — Effect of OCR fine-tuning.** Domain adaptation through OCR fine-tuning improves F1 from 0.715 to 0.738 on Online Misogyny, confirming that adapting the model to the target domain matters. This is the best text-only model in our pipeline.

**Step 3 — Effect of multimodal fusion.** The multimodal models clearly outperform all text-only variants. The best result (F1 = 0.855) combines the Step 2 fine-tuned RoBERTa with CLIP, confirming two findings: (1) visual features provide complementary signals (+6.6 points over CLIP image-only), and (2) the domain-specific text encoder helps even in multimodal settings (+0.6 points over RoBERTa base + CLIP).

**Analysis — Implicit vs explicit sexism.** The model detects 81.0% of explicit sexist examples but only 55.6% of implicit ones on EDOS — a gap of 25 points (Table 2). The hardest categories are backhanded compliments (e.g. *"She's surprisingly competent"*), condescending advice, and systemic discrimination, where predicted misogyny probability is close to zero.

| Sexism type | Samples | Recall |
|---|---|---|
| Explicit | 725 | 0.810 |
| Implicit | 245 | 0.556 |

*Table 2.* RoBERTa recall by sexism type on EDOS.

**Analysis — Gender-term bias.** The baseline false positive rate on text with no gendered terms is 1.1%, but rises to 23.8% for *female* and 18.4% for *women* (Table 3), a 22-fold increase. The model would flag many neutral or positive discussions about women as misogynistic, a bias likely inherited from training data where gendered terms appeared more often in negative contexts.

| Term | Samples | FP Rate |
|---|---|---|
| *female* | 21 | 0.238 |
| *women* | 103 | 0.184 |
| *woman* | 42 | 0.167 |
| *feminist* | 31 | 0.097 |
| No gendered term | 458 | 0.011 |

*Table 3.* False positive rates by gendered term.

**Analysis — Synthetic diagnostic benchmark.** We generated 300 controlled examples with ChatGPT (100 explicit, 100 implicit, 100 neutral/feminist) plus 50 emoji pairs to stress-test specific failure modes. The recall gap widens dramatically: 0.67 on explicit vs 0.06 on implicit — 61 points, compared to 25 on EDOS. Backhanded compliments, competence doubt, and condescending statements all reach zero recall. On neutral counterexamples, the false positive rate on women-rights-support examples reaches 35.7%. We also tested whether synthetic data could be used as training augmentation: a mixed real+synthetic approach improved synthetic implicit recall from 0.06 to 0.82, but this improvement did not transfer to real EDOS implicit sexism, where implicit recall actually decreased from 55.5% to 49.0%. This confirms that synthetic examples are useful as a diagnostic tool but do not yet capture the full complexity of real-world implicit misogyny.

**Limitations.** First, implicit misogyny remains a blind spot: the recall gap reaches 25 points on EDOS and 61 on the synthetic benchmark. Second, the model over-associates gendered terms with misogyny, flagging women-rights-support content up to 35.7% of the time. Third, synthetic augmentation improves controlled diagnostics but does not transfer to real data. Fourth, the Online Misogyny test set has just a few emojis, preventing proper evaluation of the emoji component. Finally, the multimodal model is only tested on MAMI, so generalization to other meme datasets remains unknown. More fundamentally, no existing real-world dataset combines fine-grained implicit/explicit sexism annotations with a large proportion of emoji-rich content, which is precisely why we generated synthetic examples.

## 5. Conclusion

We built a progressive pipeline for misogyny detection combining emoji-aware text encoding, domain adaptation, and multimodal fusion. Our key findings are: (1) transformer models miss implicit misogyny with a 25-point recall gap on real data and 61 on synthetic; (2) they develop strong bias toward gendered terms (FP rate 23.8% on "female" vs 1.1% baseline); (3) synthetic data helps diagnostics but does not transfer to real data; (4) emojis have limited impact on final predictions, partly because real test sets lack them; and (5) multimodal fusion with frozen CLIP reaches F1 = 0.855 on MAMI. The core challenge is model understanding, not data quantity: current transformers rely on surface-level cues rather than social context. Future work should focus on context-aware models and emoji-rich, multimodal evaluation datasets.

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